

## **Key Learnings Report**

### **Introduction**

The Raritone Summer Internship 2026 provided me with an opportunity to explore Artificial Intelligence applications in the fashion industry. Throughout the internship, I researched recommendation systems, virtual try-on technologies, fashion datasets, and modern AI models used in e-commerce platforms. This experience helped me gain both technical knowledge and practical research skills.

### **Understanding Recommendation Systems**

One of the most important learnings from this internship was understanding how recommendation systems work. I studied Content-Based Filtering, Collaborative Filtering, Hybrid Recommendation Systems, and Deep Learning-Based Recommendation Models.

Through this research, I learned how online platforms analyze user preferences, browsing history, and interactions to provide personalized product recommendations. I also gained insight into the strengths and limitations of different recommendation approaches and how they contribute to improving customer experience.

### **Knowledge of Computer Vision Applications**

The internship introduced me to the role of Computer Vision in fashion technology. I learned how AI systems process and analyze images to recognize clothing items, identify garment features, and generate virtual try-on results.

Understanding computer vision applications helped me appreciate how image processing technologies are used to create interactive and personalized shopping experiences.

### **Learning Virtual Try-On Technologies**

A major part of the internship focused on studying virtual try-on systems. I explored both 2D and 3D virtual try-on approaches and analyzed how they help users visualize clothing before making purchasing decisions.

I researched models such as VITON, CP-VTON, HR-VITON, StableVITON, and IDM-VTON. Studying these models helped me understand garment alignment, image generation, texture preservation, and the evolution of virtual try-on technology.

This learning experience provided valuable insights into how Artificial Intelligence is transforming online fashion retail.

## **Understanding Fashion Datasets**

Another important learning outcome was understanding the role of datasets in Artificial Intelligence projects. I explored datasets such as DeepFashion, DeepFashion2, FashionGen, and DressCode and learned how they are used to train recommendation systems and virtual try-on models.

This helped me understand the importance of data quality, annotations, and dataset diversity in improving AI model performance and accuracy.

## **Research and Analytical Skills**

Throughout the internship, I spent significant time reading research papers, comparing model architectures, and analyzing findings from multiple sources. This process improved my ability to understand technical content, identify important information, and evaluate different approaches objectively.

I also learned how to organize research findings, prepare comparison reports, and document technical information in a structured manner.

## **Technical Documentation Skills**

Preparing reports, documentation, workflows, and presentations helped me improve my technical writing and documentation skills. I learned how to present research findings clearly and professionally while maintaining proper organization and structure.

These skills are valuable for both academic and professional projects where effective communication of technical information is important.

## **Understanding AI Applications in Fashion**

One of the biggest takeaways from this internship was understanding how Artificial Intelligence is being applied in real-world fashion and e-commerce platforms.

The research demonstrated how recommendation systems, virtual try-on technologies, computer vision, and generative AI can work together to create more personalized and engaging shopping experiences.

This understanding provided a broader perspective on the future of AI-driven solutions in the fashion industry.

## **Conclusion**

Overall, this internship was a valuable learning experience that enhanced my knowledge of Artificial Intelligence, Deep Learning, Computer Vision, Recommendation Systems, and Virtual Try-On Technologies. It also strengthened my research, analytical thinking, problem-solving, and documentation skills.

The knowledge and experience gained during this internship will be beneficial for my future academic learning and professional development in the field of Artificial Intelligence and Machine Learning.